

Indian Institute of Technology Palakkad

Department of Mathematics

MA 5009 (Ordinary differential equations)

2024–2025 (August–November Semester)

Instructor: Dr Gopikrishnan C R

Office: 10, D03-FF (Dr. APJ Abdul Kalam Block).

Syllabus

- ▶ Review of basic concepts and solution techniques, power series solutions, properties of Legendre polynomials and Bessel functions.
- ▶ Qualitative properties of solutions : Oscillations and Sturm separation theorem, Sturm comparison theorem.
- ▶ Existence and Uniqueness Theorems for systems: Contraction mapping theorem, Peano's and Picard's theorems, Gronwall's inequality, Maximal interval of existence.
- ▶ Linear systems: The fundamental matrix, exponential of a matrix, solution to linear systems, critical points and stability.
- ▶ Nonlinear systems: Autonomous systems, phase plane analysis.
- ▶ Boundary value problems: Sturm-Liouville theory.

References

- [1] T. M. U. *Ordinary Differential Equations*. Elsevier North-Holland, 1978. ISBN 9780444002334. URL <https://books.google.co.in/books?id=7iHvAAAAMAAJ>.

Additional reading material

1. Coddington, E. A. and Levinson, N., *Theory of Ordinary Differential Equations*, McGraw-Hill, 1972, ISBN-10: 9780070992566.